Evaluating the Effect of Play-Action Passing on Defensive Disruption

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Abstract

Play-action has become a key part of modern NFL offenses, yet most research rarely ever explains why it works. This study analyzed how play-action relates to the defense's reaction at the snap, and whether that reaction shows up in pressure, offensive efficiency, and overall defensive disruption. Using 7,368 dropbacks from weeks 1 through 9 of the 2022 NFL season, I applied multiple linear regression, logistic regression, and situational matching on the down, distance, field position, and score. Across the league pass rushers got off the line 0.22 seconds slower on play-action dropbacks. Play-action also gained 0.11 EPA per dropback, with most of that gain coming on first down. However, play-action was associated with higher odds of pressure, and the slower get-offs had no direct link to EPA when tested. This suggests that though play-action does change how the pass rush starts, this slower start alone does not explain its value, and there are more likely many more factors at play that are causing the success of play action.

Introduction

Football has changed drastically since its inception, as rule changes and adaptations have evolved the game to something unrecognizable from the beginning years of football. Over the past two decades, football has shifted away from run-heavy offenses toward more explosive passing schemes, usually from the shotgun formation. However, in recent years, offenses have leaned more and more on a run fake known as "play-action" to become even more efficient. Eager and Seth (2023) define play-action passes as plays where the offense fakes a handoff to a ball carrier before the quarterback drops back to attempt a throw. The goal of this is to stop the front seven defenders from rushing the passer, and draw linebackers forward against the fake

run, opening space behind them to make easier throws in the intermediate part of the field. While it is widely accepted that play-action increases offensive production (Sharp, 2019), existing research never really explains the "why". Most of the studies stop at basic stats such as yardage or Expected Points Added (EPA), which gloss over the critical first second after the snap, where the actual deception happens. When just looking at the final yardage or EPA of a play, it is impossible to tell if play-action worked because of actual hesitation or confusion by the defense, or because offenses usually call it in advantageous situations. Calling plays for an offense comes down to a makeshift chess match between the offensive coordinator and the defense. Football has even been looked at through game theory by comparing it to the matching pennies game (McGarrity & Linnen, 2010). In this setup, the offense picks between a run and a pass, while the defense commits its personnel to stop one or the other. This is a great example of how important it is for offenses to stay unpredictable. Even so, NFL coaches still often leave value on the table on early downs. Using Generalized Additive Models, Wilk (2026) found that throwing the ball is the better analytical choice over 80% of the time on first down, yet offenses still run the ball on about half of their first downs. This gap between what coaches actually do and what the analytics say they should do is where play-action proves to be extremely effective. On these early downs, defenses naturally expect the offense to run because that is what coaches have traditionally called for a majority of the history of the NFL. When an offense then fakes the handoff, they not only get to take advantage of the defense expecting a run, but also get to pass the ball. However, a final box score number cannot tell you if a linebacker hesitated for a second or took a step in the wrong direction due to the play fake, which is the exact physical reaction that makes play-action work in the first place. To measure what happens in that first second after the snap, new research has turned to player tracking data from RFID chips in players' shoulder pads. Eager and Seth

(2023) built a linebacker metric around “bite distance”, the amount a player moves toward the line of scrimmage on play-action, and Eager et al. (2022) combined charting and tracking data to better evaluate pass rushers, linebackers, and receivers. Attrah (2026) also found a significant mismatch between orientation and movement for defensive players, which helps explain the physical cost of a fake. When a linebacker bites on play-action, they have to stop their forward momentum and flip their hips to get back to their coverage responsibility. My approach also measures the defense’s reaction, but it focuses more on the pass rush, and it connects that reaction to pressure and EPA. While plenty of work has been done on play-action, most studies do a great job of proving that it is effective, but fall short of explaining the actual “why,” which is extremely important to the teaching part of football that helps evolve the game at all levels. One of my goals for this research is to help influence offensive game plans at the high school and college level. It usually takes years for NFL trends to trickle down to lower levels of football, and I hope that more practical research like this could help expedite that process.

This paper asks whether play-action is associated with a different defensive reaction at the snap, and whether that difference shows up in pressure and in offensive efficiency via EPA. I expected play-action to slow the pass rush, that a slower rush would reduce pressure, and that this would help explain any gain in EPA. To examine this relationship, I used data from the NFL Big Data Bowl (National Football League, 2025) and applied regression and situational matching to 7,368 dropbacks. Because play calling is not random, every result is an association and not a causal effect.

Methods

Data and Sample

To examine this relationship, I used the NFL Big Data Bowl data on Kaggle (National Football League, 2025), which covers weeks 1 through 9 of the 2022 season and comes from the 2025 competition. The data is open-access for public research, so I can analyze it and share my findings without ethical or privacy concerns. I joined four files provided by the Big Data Bowl: plays.csv, player_play.csv, players.csv, and games.csv, using gameId and playId. Everything was done in R (R Core Team, 2026), mostly with the tidyverse (Wickham et al., 2019) package. I kept dropbacks on downs one to three with ten or fewer yards to go. I did this because the run has to be a realistic option for a fake handoff to potentially work, and on third and long the defense is not really worried about the run. I also removed quarterback spikes, plays nullified by a penalty, overtime, and garbage time, because these scenarios can completely change how a defense normally plays and would likely just cause noise. I defined garbage time as the fourth quarter with a score margin above sixteen points in either direction, since this would make it more than a two-score game. This left me with 7,368 plays, made up of 5,243 standard dropbacks and 2,125 play-action dropbacks. There were 51 plays missing get-off time and 2 missing pressure for an unknown reason. I handled this by only dropping them out of the models that needed those variables.

Variables

The main explanatory variable is a true-or-false indicator for whether the offense used play-action. To measure how the defense reacted, I used pass rusher get-off time, the number of seconds a pass rusher takes to cross the line of scrimmage after the snap. This is recorded per player, so I averaged it across each play's initial pass rushers, and a higher number means a slower rush. A play counted as pressured if any initial rusher caused a pressure, which the data flags as a peak pressure probability of at least 0.75. I measured offensive efficiency with EPA,

which tracks the change in expected points before and after the snap, and I also used yards per dropback. The controls I used are down, yards to go, field position, and score differential from the offense's point of view. All four are set before the snap, because controlling for something that happens after the snap could hide part of what play-action is actually doing and hurt the accuracy of my results.

Regression Models

A raw comparison between the two play types would not be fair. In my sample, 59.9% of play-action dropbacks came on first down, compared to 33.5% of standard dropbacks, and defenses rush differently on different downs. So I ran regressions first, then used matching as a second approach.

Get-off time: To see whether play-action goes with a slower rush, I ran this on all plays:

$$GetOff = \beta0 + \beta1(PlayAction) + \beta2(Down2) + \beta3(Down3) + \beta4(YardsToGo) + \beta5(AbsoluteYardline) + \beta6(ScoreDiff) + \varepsilon$$

β1 is the difference between a play-action play and a standard dropback after accounting for everything else, and Down2 and Down3 compare second and third down to first. I reran this with the number of rushers added, with only defensive linemen, and separately for each down.

EPA and yards: For offensive efficiency, the right side stayed the same:

$$EPA = \beta0 + \beta1(PlayAction) + \beta2(Down2) + \beta3(Down3) + \beta4(YardsToGo) + \beta5(AbsoluteYardline) + \beta6(ScoreDiff) + \varepsilon$$

$$\text{Yards} = \beta_0 + \beta_1(\text{PlayAction}) + \beta_2(\text{Down2}) + \beta_3(\text{Down3}) + \beta_4(\text{YardsToGo}) + \beta_5(\text{AbsoluteYardline}) + \beta_6(\text{ScoreDiff}) + \varepsilon$$

Get-off and EPA: To see if a slower rush goes with more EPA, I put get-off in the EPA model:

$$\text{EPA} = \beta_0 + \beta_1(\text{GetOff}) + \beta_2(\text{PlayAction}) + \beta_3(\text{Down2}) + \beta_4(\text{Down3}) + \beta_5(\text{YardsToGo}) + \beta_6(\text{AbsoluteYardline}) + \beta_7(\text{ScoreDiff}) + \varepsilon$$

Pressure: Pressure is binary, so I used a logistic regression, where p is the probability of pressure:

$$\log(p / (1 - p)) = \beta_0 + \beta_1(\text{PlayAction}) + \beta_2(\text{Down2}) + \beta_3(\text{Down3}) + \beta_4(\text{YardsToGo}) + \beta_5(\text{AbsoluteYardline}) + \beta_6(\text{ScoreDiff})$$

I report these as odds ratios, where anything above 1 means higher odds of pressure. To confirm that a slower start goes with less pressure, I added get-off:

$$\log(p / (1 - p)) = \beta_0 + \beta_1(\text{GetOff}) + \beta_2(\text{PlayAction}) + \beta_3(\text{Down2}) + \beta_4(\text{Down3}) + \beta_5(\text{YardsToGo}) + \beta_6(\text{AbsoluteYardline}) + \beta_7(\text{ScoreDiff})$$

Situational Matching

To see the true value of play-action, I wanted to directly compare play-action plays to standard dropbacks from as similar spots as possible. So I used exact matching on grouped versions of my four controls. Down stayed as first, second, or third. I grouped distance to first down into 1 to 3, 4 to 6, and 7 to 10 yards, field position into four groups with the same number of plays, and score into down 9 or more, down 1 to 8, tied or up 1 to 8, and up 9 or more. Every combination made one cell, and I got rid of any cells that did not have both play types. The matched sample ended up having 6,853 plays, with 2,118 play-action plays and 4,735 standard dropback plays.

Within each of the cells, play-action plays got a weight of 1, and standard dropbacks got the number of play-action plays divided by the number of standard dropbacks, so both play types counted the same in every cell. I then ran the same regressions on these matched plays with the weights applied to them, which can be seen in the table below.

Table 1. Situation Balance Before and After Matching.

Measure	***Before: Standard***	***Before: Play-Action***	***After: Standard***	***After: Play-Action***
First-down share	33.5%	59.9%	60.1%	60.1%
Yards to go	7.72	8.27	8.41	8.29
Field position	60.6	60.9	60.8	60.9
Score margin	−2.41	−1.28	−1.17	−1.32

After-matching values are weighted. Weeks 1 to 9 of the 2022 NFL season.

As Table 1 shows, the first-down gap is gone, and score margin and yards to go got much closer, proving that the matching worked to make them comparable before I tested with them.

Where the EPA Gain Shows Up

Once I confirmed an EPA gain, I wanted to know exactly where it was coming from. So I ran the matched EPA model on first down and on second and third down combined, then tested the difference with an interaction:

$$EPA = \beta0 + \beta1(PlayAction) + \beta2(FirstDown) + \beta3(PlayAction \times FirstDown) + \beta4(YardsToGo) + \beta5(AbsoluteYardline) + \beta6(ScoreDiff) + \varepsilon$$

Pressure also got the same treatment:

$$EPA = \beta0 + \beta1(PlayAction) + \beta2(Pressure) + \beta3(PlayAction \times Pressure) + \beta4(Down2) + \beta5(Down3) + \beta6(YardsToGo) + \beta7(AbsoluteYardline) + \beta8(ScoreDiff) + \varepsilon$$

I also added pressure to the matched EPA model to see the impact it had on an offense, and ran the model by distance and for third-and-short, meaning just three yards or less to go. I decided to split the plays this way after seeing my first results, so these should be seen as exploratory and not firm conclusions. The pressure split also needs some caution, since pressure happens after the snap and play-action itself may be affecting it.

Checking the Models

When using a linear regression, it assumes constant error variance, or homoscedasticity, which I checked with the Breusch-Pagan test. It was significant for both the get-off model ($BP = 230.9$, $p < .001$) and the EPA model ($BP = 211.2$, $p < .001$), so the variance was not constant in either of the models. The estimates are fine, but the usual standard errors can't be fully trusted due to this. Residual and Q-Q plots showed a long lower tail for EPA, which makes sense due to the impact sacks and turnovers have. Variance Inflation Factors were all below 1.3, so multicollinearity was not a problem.

Independence was another concern, since plays from the same game share the same teams, weather, and other small factors, so treating every play as independent can make intervals look more certain than they should be and needs to be addressed. To deal with both this and the uneven variance, I used robust standard errors clustered by game across all 136 games. It is also important to note that all 95% confidence intervals in this paper use these standard errors unless I say otherwise. The estimates stayed the same, but most intervals did get wider, so this did change how sure I can be about the results of this study. Clustering does not account for the same

defense showing up in multiple games, so I also reran the get-off and EPA models with random effects for the game and the defense. This uses the same regression idea as before, but gives each game and each defense its own starting point, since some defenses are just better because they are faster, or worse because they are slower than others:

$$Y_{ijk} = \beta_0 + \beta_1(\text{PlayAction}) + \beta_2(\text{Down2}) + \beta_3(\text{Down3}) + \beta_4(\text{YardsToGo}) + \beta_5(\text{AbsoluteYardline}) + \beta_6(\text{ScoreDiff}) + u_j + v_k + \varepsilon_{ijk}$$

In this, i is the play, uj is the adjustment for game j, and vk is the adjustment for defense k. EPA also has some extreme plays, so I reran the main EPA models after dropping both the top and bottom 1% and 2.5% of plays to see if a few huge plays were the reason for the gain. The following section presents all of the main findings.

Results

Pass Rush Get-Off

As Figure 1 shows, the typical play-action get-off was 1.17 seconds compared to 0.90 seconds on standard dropbacks, and the entire top 50% of play-action plays sits above standard dropbacks in get-off time.

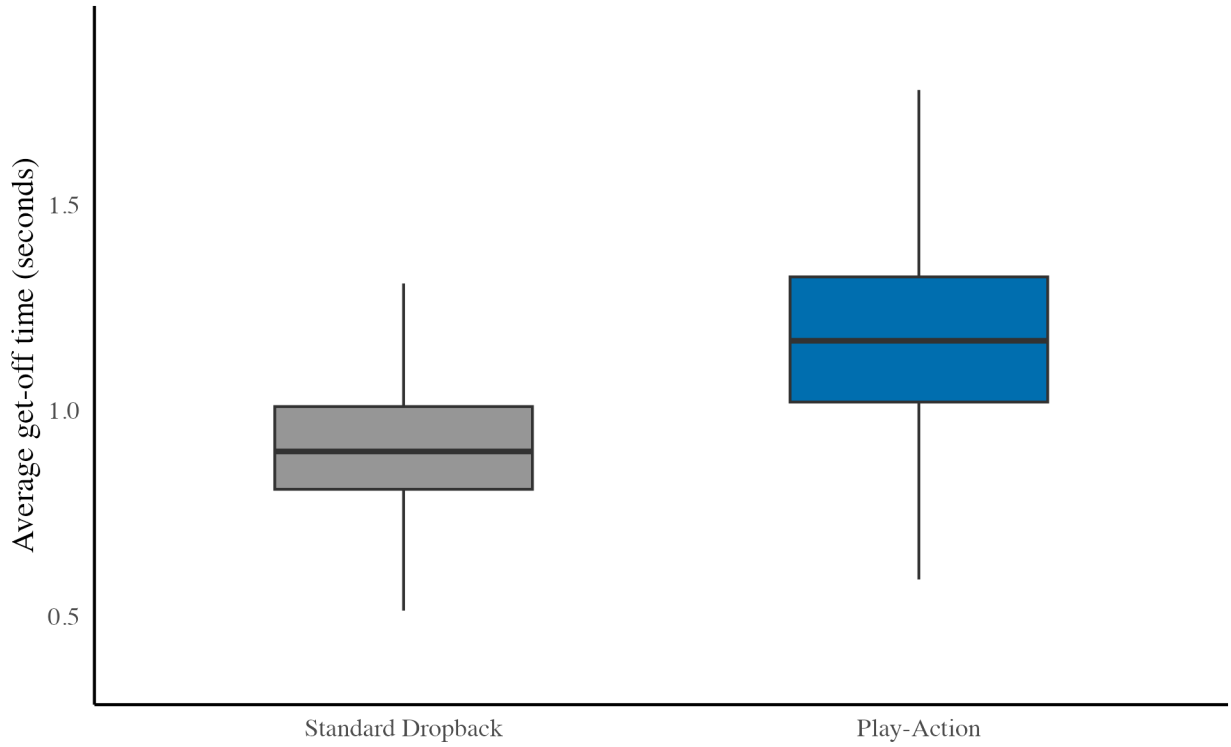


Figure 1. Pass Rushers Get Off the Line Slower on Play-Action.

Average get-off time per play by play type. Boxes show the middle 50% of plays and lines show the median. Extreme values are not shown to keep the boxes readable, but they were included in the model. Weeks 1 to 9 of the 2022 NFL season.

This gap held up after controlling for situation. In the regression, rushers got off the line 0.226 seconds slower on play-action, with a 95% confidence interval from 0.213 to 0.24. After matching, the difference was just 0.221 seconds, with an interval from 0.207 to 0.235. Overall, rushers were getting off 25% slower during play-action dropbacks than on a standard dropback. This is a huge difference in terms of rushing the passer, and both methods tell nearly the same story (Table 2).

Table 2. Play-Action Effect on Get-Off, EPA, and Yards by Method.

Outcome	Regression	Regression 95% CI	Matched	Matched 95% CI
Get-off time (seconds)	0.226	0.213 to 0.240	0.221	0.207 to 0.235
EPA per dropback	0.087	0.011 to 0.163	0.111	0.035 to 0.188
Yards per dropback	1.342	0.780 to 1.904	1.462	0.892 to 2.033

Table 2. Play-Action Effect by Method.

Regression uses all 7,368 plays and matched models use 6,853 weighted plays, with fewer plays for get-off because of missing values. Both control for down, distance, field position, and score. Intervals use standard errors clustered by the game.

This also checks out from a football standpoint. Compared to first down, rushers got off about 0.04 seconds faster on second down and about 0.12 seconds faster on third down, which is likely because defenses are usually expecting a pass on later downs. The play-action slowdown stayed constant throughout all of the downs, at 0.226 seconds on first down, 0.225 on second down, and 0.233 on third down. It also showed up for all 32 of the teams' defenses, meaning the effect is seen league-wide. Adding the number of rushers barely changed the result down to 0.222 seconds, and looking at only defensive linemen made it slightly larger, at 0.240 seconds. So the slowdown does not seem to be caused by who is rushing, and is more likely being impacted by other factors in the play that are difficult to accurately measure.

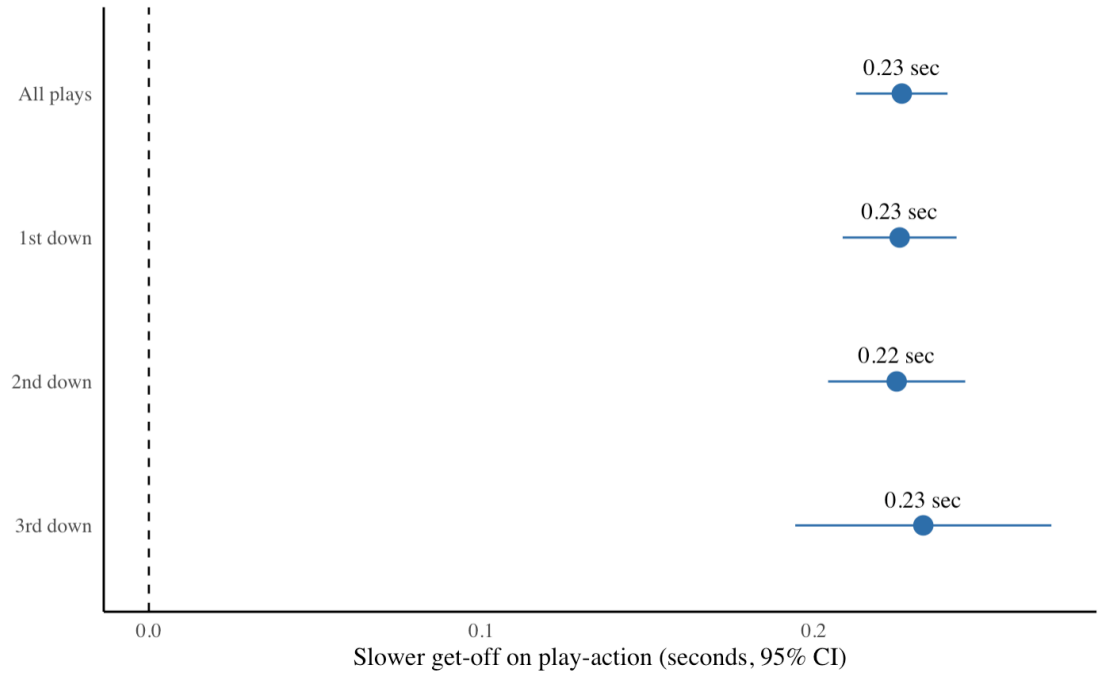


Figure 2. Play-Action Slows the Pass Rush on Every Down.

Estimated difference in get-off time between play-action and standard dropbacks, with 95% confidence intervals. Weeks 1 to 9 of the 2022 NFL season.

Pressure

To see if this slower pass rush get-off mattered, I also looked at pressure, and I found that slower get-offs did go hand and hand with less pressure. Each 0.1 seconds of slower get-off for pass rushers was associated with around 10% lower odds of pressure, with an odds ratio of 0.9. However, play-action itself was also associated with higher odds of pressure, with an odds ratio of 1.22 and a 95% confidence interval from 1.10 to 1.36. So even though rushers started slower on play-action, the quarterback was still pressured more often. This may be due to the fact that most play-action plays are longer developing, or put offensive linemen in difficult pass blocking situations as they first fake run blocking. Though it is not something that can be explored with

the current data available. This is however something that the raw numbers hide, as pressure rates looked almost even at 32.8% on standard dropbacks and 31.9% on play-action. But once the plays were matched, the rates were 27.6% for standard dropbacks and 31.9% for play-action. This is most likely due to the high percentage of play-action that occurs on first down. Pressure is also tied closely to the down, as the odds of pressure on third down were about 2.5 times the odds on first down. This makes sense because as mentioned above teams are more prepared for the pass on third down, and call more exotic defenses to confuse offensive playcallers. Figure 3 shows this split by down, as play-action drew more pressure on first and second down but less on third down. With get-off added to the model, the play-action odds ratio rose to 1.54, with an interval from 1.36 to 1.75. This means that for two plays with the same get-off, the play-action play had a much higher odds of facing pressure from the defense.

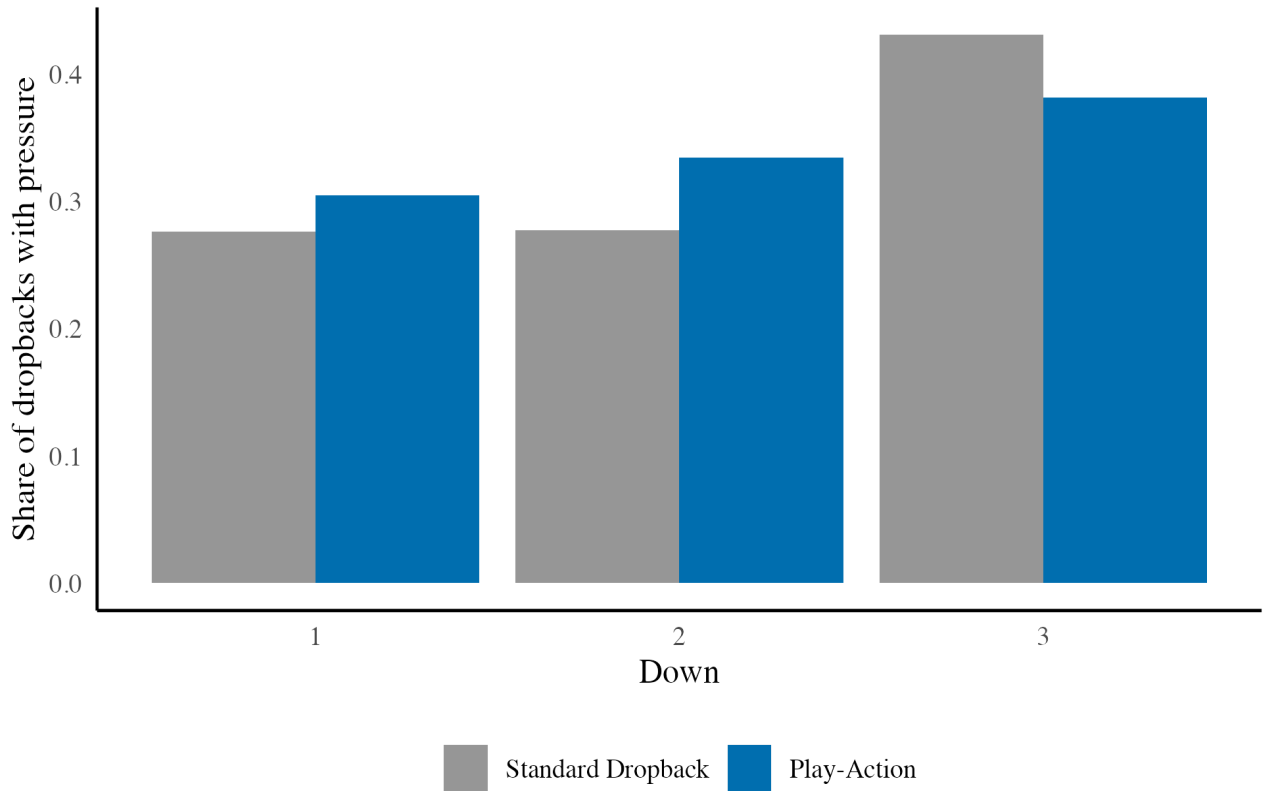


Figure 3. Play-Action Draws More Pressure on 1st and 2nd Down but Less on 3rd Down.

Share of the dropbacks with pressure by down and play type. Weeks 1 to 9 of the 2022 NFL season.

Offensive Efficiency

The next question I had is whether play-action is actually adding expected points. On average, play-action dropbacks had an EPA of 0.084, compared to −0.047 for standard dropbacks. Showing a clear advantage to calling play-action. In the regression, play-action gained 0.087 EPA per dropback, with a confidence interval from 0.011 to 0.163. After the matching, the gain in EPA was 0.111, with an interval from 0.035 to 0.188. So an offense does gain a small amount of expected points each time it uses play-action. Though the intervals do come extremely close to

including zero, they do stay above that key threshold. Yards also show a very similar thing, as play-action gained 1.34 more yards per dropback in the regression and gained 1.46 more after the matching (Table 2). I then looked at whether the slower rush is correlated with more offensive EPA. When I ran the regression test, get-off had a coefficient of just 0.011 with a p-value of .91, so there was no statistically significant link between the two. Pressure, on the other hand, is extremely costly to an offense. A pressured dropback had a loss of about 0.53 EPA, with an interval from −0.61 to −0.46. However, that lower pressure cannot explain the play-action gain. When I added pressure to the matched EPA model, the play-action gain went up from 0.111 to 0.134 instead of shrinking like I originally expected. Meaning play-action is not helping by stopping pressure, it is likely providing its advantages to an offense another way.

Where the EPA Gain Shows Up

EPA gain did not come from each of the downs equally. Throughout the NFL offenses pass on around 50% of their first downs, which makes first down basically just a coin flip between run and pass. This means first down is when a defense is most likely to expect the run, and therefore where a fake handoff should be at its most effective in disrupting the defense, leading to more offensive efficiency. The results show this clearly, as the EPA gain on play-action dropbacks was 0.193 on first down, with an interval from 0.086 to 0.299, and −0.016 on second and third down combined, with an interval from −0.135 to 0.103 (Figure 4). It is also important to note that the first-down gap of 0.203 EPA was statistically significant, which then shows that the overall gain of offensive efficiency via play-action is mostly coming from first down plays. Which suggests that offenses should be using play-action even more on first down than they already are.

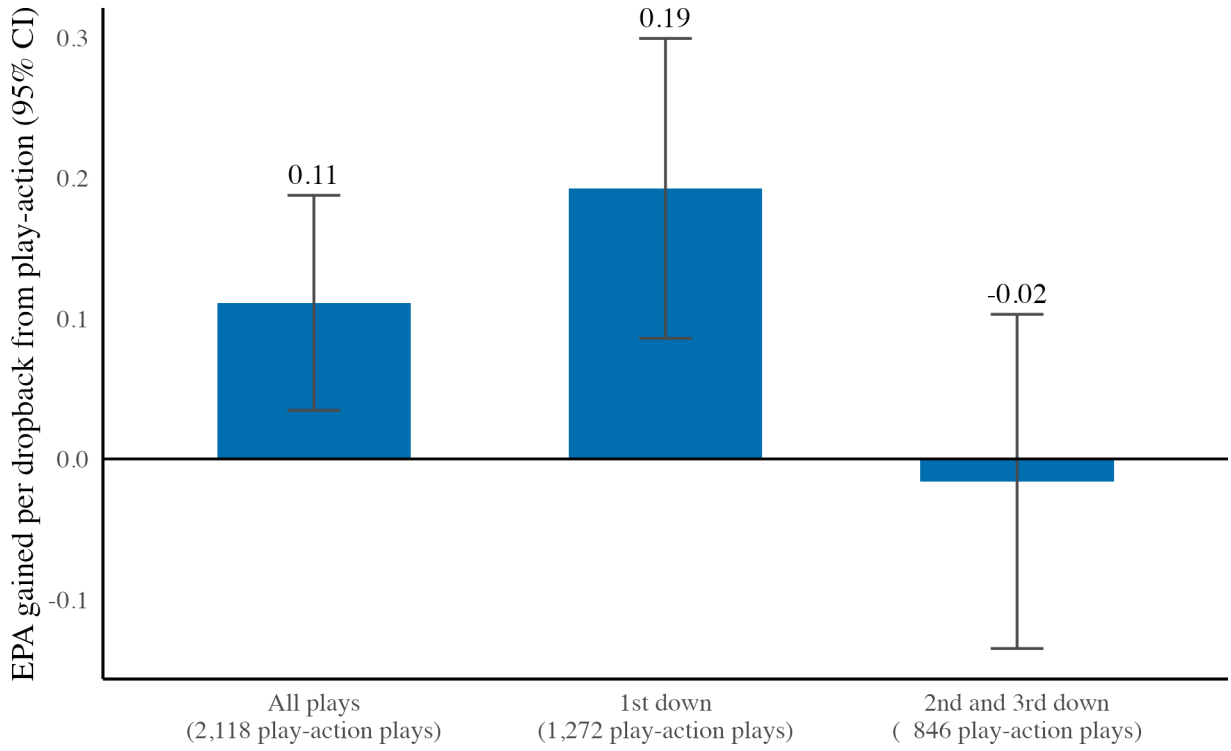


Figure 4. Play-Action Adds EPA on First Down but Shows No Clear Gain on Later Downs.

Estimated EPA gained per dropback from play-action among matched plays, with 95% confidence intervals. Weeks 1 to 9 of the 2022 NFL season.

Pressure also showed a pretty similar pattern; on plays where the quarterback was not under pressure, play-action gained 0.211 EPA, with an interval from 0.122 to 0.300. But when he was pressured, the estimate was −0.045, with an interval from −0.192 to 0.101 (Figure 5). This clear difference of −0.250 EPA was statistically significant, so play-action seems to pay off when the quarterback stays clean, but that advantage goes away once pressure gets there. Making it clear that play-action is successful but does carry some serious flaws if not run correctly.

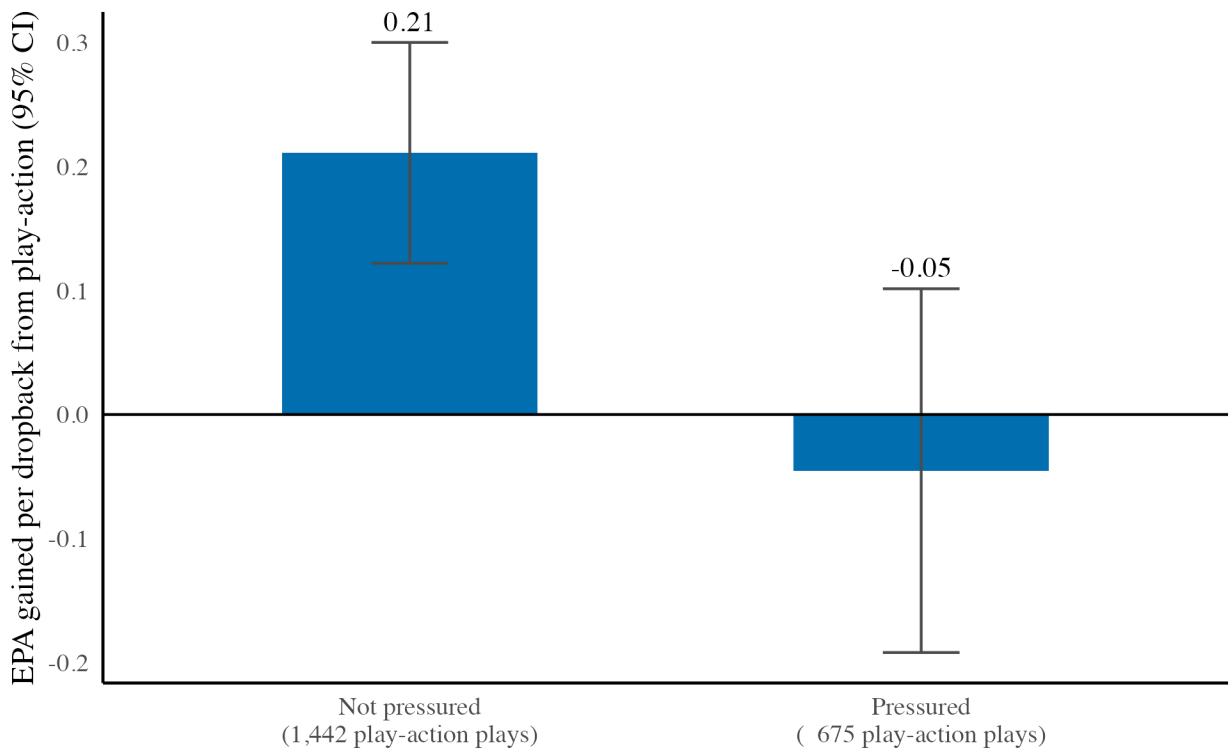


Figure 5. Play-Action Adds EPA, But Only When the Quarterback Is Kept Clean From Pressure.

Estimated EPA gained per dropback from play-action among matched plays, with 95% confidence intervals. Weeks 1 to 9 of the 2022 NFL season.

However, looking at the different distances gave more mixed results. Play-action gained 0.413 EPA at 1 to 3 yards to go and 0.121 at 7 to 10 yards, but lost 0.230 at 4 to 6 yards. I believe one reason for this is that running plays are common at short distances, so a run fake is a lot more believable to the defense. The 7 to 10 yard range is made up of a lot of first down plays, with 1,231 of its 1,656 play-action plays coming on first down. As I mentioned above, first down is basically a coin flip between a run and a pass, so it makes sense that play-action would be effective there too. The 4 to 6 yard range on the other hand is a little harder to explain, since most of those plays came on second down, where offenses still will run the ball about half the time. Because of this, and the smaller sample sizes, I would not read too much into that 4 to 6

yard result. I also wanted to see if third-and-short could be just as effective as first down, since a run is a realistic option there too. But unfortunately there were only 83 play-action plays that fit that situation, and the estimated gain of 0.07 EPA had an interval from −0.40 to 0.54, making the results unreliable. This means that question is still unanswered and would need more data to answer reliably

Robustness Checks

Since some of my results were pretty close to zero, I wanted to make sure that they would still hold up if I ran things a little differently. One way I did this was by adding random effects for the game and the defense. When I did this, the results barely changed at all. Pass rushers still got off the line 0.225 seconds slower on play-action than standard dropbacks, with a 95% confidence interval from 0.215 to 0.234. Which is almost the exact same as the 0.226 seconds from my main regression. Play-action also still gained 0.087 EPA per dropback, with an interval from 0.003 to 0.170, which exactly matches the 0.087 from the main regression I ran. This helps show that the results were not being caused by a few specific games or defenses, and are more trustworthy. I did a similar test of EPA gain, as I wanted to know if a few huge plays or really negative plays were behind the EPA gain results. After dropping the top and bottom 1% of plays, the matched gain went down from 0.111 to 0.089, with an interval from 0.029 to 0.149. When I dropped the top and bottom 2.5%, it went down to 0.078, with an interval from 0.023 to 0.134. So big plays and turnovers do push the number up some, but the gain is still there without them. It is also important to mention that these big plays are not really outliers in the sense of classic data examples, as they are an important product of using play-action and the risk involved with calling it. Overall, while the get-off difference stayed almost exactly the same no matter what I changed, the EPA effect was less certain but still held up pretty well.

Discussion

The main goal of this paper was to see whether play-action changes what the defense does at the snap, and whether that change shows up in pressure and offensive efficiency. The clearest finding I discovered was that pass rushers got off the line about 0.25 seconds slower on play-action than they do on a standard dropback. This held up in every version of the model I ran, on every down, and for all 32 teams defenses. Play-action was also associated with a small gain in teams offensive EPA and yards, with a big portion of that EPA gain coming on first down and on plays where the quarterback was not pressured by the defense. However, one thing I expected to see was that play-action would slow the pass rush, and that slower rush would then lead to less pressure, and that less pressure would explain the gain in offensive efficiency when using play-action. While the first two steps were correct and supported through my analysis, play-action was actually associated with higher rates of pressure than standard dropbacks, and a slower pass rush get-off had no direct link to offensive EPA.

What the Results Mean

The pass rush get-off result does support the idea that play-action causes some hesitation in the defense right at the snap, especially for the pass rushers. This is consistent from the analysis McGarrity and Linnen (2010) conducted, as a defense that has to respect the threat of the run cannot fully commit to rushing the passer. The time lost deciphering between a run and play-action is a lot of time at the line of scrimmage, and in a league where most passes come out in under three seconds, it is a real advantage for an offense. This is also a textbook example of the kind of defensive disruption that box score stats cannot truly show, which was one of the main gaps this project wanted to address from the start. The defensive pressure rate result was

the most surprising part of this study. This was because even though rushers started slower during play-action plays, quarterbacks were still pressured at a higher rate. This could be because play-action plays take longer to develop and therefore give more time for defenses to pressure the quarterback. Another factor is that offensive linemen also have to sell the run block first, which may put them in worse spots to pass protect, therefore negating any advantage the slow get-off may provide the offense. I was not able to test either of these ideas with the data I had, but they would most likely be able to help explain why a slower start does not lead to less pressure like you would expect. This also helps explain why the EPA gain did not shrink when I added pressure to the model. If play-action was helping offenses by protecting the quarterback, the gain should have gone away once pressure was accounted for in the model, but it actually went up, which was unexpected. This suggests that the value of play-action may also be coming from somewhere else, like the players in the second level of the defense in coverage. If linebackers are stepping up to defend the suspected run, the space behind them will open up and cause easy passing lanes for the offense. This also matches the idea of "bite distance" from Eager and Seth (2023) and the cost of changing direction found by Attrah (2026). Because I did not measure linebacker movement or coverages in this project, this remains an open question, but it is most likely a major unexplored factor that could help explain why play-action is still effective.

The results from first down line up well with what I expected before this research. In this dataset, offenses passed on around half of their first downs, which makes it the down where a defense has the hardest time guessing what is coming. So as mentioned above, that is exactly where play-action gained the most EPA, at about 0.19 per dropback. On second and third down combined, there was no clear gain in EPA. Which helps me agree even more with the idea from

Wilk (2026) that most teams are still leaving value on the table on early downs by not calling more play-action plays.

Who This Helps

For coaches, the main takeaway from these results should be that play-action is still severely underutilized in many modern offenses, and should continue to be implemented. It works best when the defense truly believes a run could be coming, such as on first down, and can add value to an offense that can properly protect their quarterback from pressure when it is called. For fans, this project helps explain the stark increase in play-action seen across the NFL. Showing that play-action disrupts the defenses timing, and leads to a more efficient offense.

Limitations

There are several limitations that should be noted for this research. The biggest is that this study is observational. Play calling is not random, and coaches choose their play calls based on things I could not see in the data, such as personnel, the defensive look, what they have seen on film from previous games, and how much success the run game has had that day. Because of this, every result that is observed in this paper is an association and not a direct causal effect. The data also only covers nine weeks of just one NFL season. This means that my results may not hold up when looking at other seasons, especially as defenses begin to adjust to how often teams are using play-action and counter their offensive counterparts. If future research is able to add more seasons that would certainly help, and would also help provide more plays in the specific situation like third-and-short, which only had 83 play-action plays and therefore could not be reliably researched. There are also some limits in how some of the variables were measured. Pass rush get-off time is measured by being averaged across all the initial pass rushers on a specific

play, so it does not really show if individual players are hesitating. Measuring pressure also had some limitations, as it also was only counted from the initial rushers, and because it happens after the snap, the pressure split in this paper can only be descriptive. I was also not able to filter out trick plays or run-pass options, so some plays in the data may not behave like normal dropbacks and could cause some outlier results. Plays within a game are also not fully independent, since they are being run by the same teams in the same game. To account for this issue, I clustered the standard errors by game and checked the results with random effects for the game and the defense, and thankfully the main findings held constant. This helps reduce the problem, but does not remove it completely, so it is still something to keep in mind when looking at the results. The scope of this project was also limited to just dropbacks, as I did not include running plays. Because of this, I was not able to look at whether a successful play-action offense also helps the run game, which was one of my original goals I talked about. Lastly, the results from this study come from the NFL. It is important to note that high school and college defenses are built differently and have very different talent levels than the pro's, so results may not translate directly, if at all to these other levels.

Ethical Considerations

The data I used in this project is publicly available and was released by the NFL for research purposes, so there are no privacy concerns in using it for this project. However, it is important to note that the data includes individual player information and tracking data, so I only reported results at the team and league wide level, and did not look at any individual players. I also used an AI tool, Claude (Anthropic, 2026), to help write and debug some of the R code. I ran all of the code myself and checked the results against the output to ensure they were what I was looking

for, but AI-written code can still contain mistakes or choices I did not fully think through, so my results are only as reliable as that code.

Future Research

The most important next step I would want to take is using the player tracking data to directly measure what defenders do after the snap, similar to Eager and Seth (2023). This would help show whether play-action's value really comes from defensive hesitation, from opening up throwing windows, or other things my current results cannot confirm yet. Future studies should also look into whether the 4 to 6 yard interval result is real, or just a product of a small sample. I believe it is the latter, but an expanded dataset would be the only way to confirm this hypothesis. Adding more data would also hopefully make it possible to finally answer the third-and-short question, and to see whether these trends stay constant or change as play-action continues to grow in usage across the league.

Conclusion

Play-action has become a bigger part of NFL offenses every year due to the clear advantages to offensive efficiency it provides when used, but there has been very little research on the "why" it works. The research in this project found that play-action clearly changes how the pass rush starts, as rushers get off the line about a quarter of a second slower than they do during standard dropbacks. It is also associated with a small gain in offensive efficiency, mostly on first down plays, and only when the quarterback is not pressured by the defense. However, the slower pass rush get-off does not lead to less pressure, and does not directly explain the offensive EPA gain by calling play-action. This shows that the slower start to the pass rush is only one part of the story, and that the rest of play-action's value most likely comes from what happens in the second

level of the defense after the run fake. Solving that next piece of the puzzle is crucial to understanding what makes play-action so popular and effective in the NFL, and will make play-action easier to teach and use at every level of football.

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Appendix

All of the data and code for this project are available on GitHub at:

https://github.com/cjgeier/DA-401-CJ-Geier. The raw data comes from the NFL Big Data Bowl 2025 on Kaggle.

Table A1. Model Diagnostics.

Check	Get-Off Model	EPA Model
Breusch-Pagan statistic	230.9 ($p < .001$)	211.2 ($p < .001$)
Highest adjusted VIF	1.26	1.26

Table A2. Play-Action Effect With Classic and Clustered 95% Confidence Intervals.

Outcome	Method	Estimate	Classic CI	Clustered CI
Get-off (seconds)	Regression	0.226	0.217 to 0.236	0.213 to 0.240
Get-off (seconds)	Matched	0.221	0.212 to 0.231	0.207 to 0.235
EPA per dropback	Regression	0.087	0.003 to 0.171	0.011 to 0.163
EPA per dropback	Matched	0.111	0.044 to 0.178	0.035 to 0.188
Yards per dropback	Regression	1.342	0.805 to 1.879	0.780 to 1.904
Yards per dropback	Matched	1.462	0.985 to 1.939	0.892 to 2.033